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DTCS–403

DIPLOMA (CSE)
(FOURTH SEMESTER)
MID SEMESTER
EXAMINATION, March, 2024

COMPUTER NETWORKS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What do you understand by Computer Networks ? Explain its applications. Explain the benefits of computer networks in detail. (CO1)

OR

(b) Explain the significance of Switching. Explain the working and significant difference between packet switching and circuit switching. (CO1)

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2. (a) Explain different types of classification of Computer Networks. Classify the networks based on the geography. Compare and contrast all of its types of Networks. (CO1)

OR

(b) What is Communication System ? Explain the components of the communication system in detail. (CO1)

3. (a) How peers' architecture is different from client-server architecture ? Compare the performance for both architectures having n number of active hosts for data frame of size F bits in terms of delay. What is Peer architecture, explain with the help of bit torrent and a diagram. (CO1)

OR

(b) A : Suppose , we have a big single network having IP Address 200.1.2.0. Divide this network into 4 subnets. (CO1/CO2)

B : What is the sub network address for a host with the IP address 200.10.5.68/28 ?

4. (a) What is the significance of topologies ? What are the different types of topologies ? Explain three types of topologies with their advantages and disadvantages (other than Mesh and Hybrid). (CO2)

OR

(b) Differentiate between Hybrid and Mesh Topology with their advantages and disadvantages. (CO2)

5. (a) Write the short notes on the following : (CO2)

(i) Switches
(ii) Routers
(iii) Modern
(iv) Hubs
(v) Gateways

OR

(b) What is the significance of Bridges ? Explain the different types of Bridges. (CO2)

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